



LUNDS
UNIVERSITET

Tentamensskrivning
Topology
Monday May 25, 2009
Time: 08.00–13.00

Matematikcentrum
Matematik NF

No aids allowed. Use the distributed paper sheets and write only on one side, with at most one problem per page. Fill in the cover sheet completely and initialize every sheet. Write legibly (in Swedish or English). Motivate your conclusions clearly and concisely; draw a picture if that helps to clarify your argument.

Test results: Will be posted on Tuesday May 26, before 17.00.

Oral exams: On Thursday May 28 and Friday May 29. State your preferred day and AM or PM on the cover sheet of your test, with at least two options.

1. Let

$$\mathcal{E} = \left\{ \sum_{k=0}^m a_k e^{kx} : a_k \in \mathbf{R}, m \in \mathbf{Z}_+ \right\}$$

Show that $\mathcal{E}$ is dense in $\mathcal{C}([-1, 1], \mathbf{R})$.

2. Let $\mathcal{T}$ be a family of sets $U \subseteq \mathbf{R}^3$ such that

$$U = \{ (x_1, x_2, x_3) : (x_1, x_2) \in V, x_3 = r \} = V \times \{ r \}$$

where V is open in the usual topology on $\mathbf{R}^2$ and $r \in \mathbf{R}$.

- Show that these sets form a base for a topology on $\mathbf{R}^3$.
 - Show that this topology is larger than the standard topology on $\mathbf{R}^3$.
 - Which are the functions that are continuous with respect to this topology?
 - Show that this topology is not separable.
 - What are the relative topologies inherited by $\mathcal{T}$ on the plane $\mathbf{R}^2 \times 0$ and on the line $L = \{ (0, 0) \} \times \mathbf{R} = \{ (0, 0, x_3) : x_3 \in \mathbf{R} \}$?
3. A mapping $f : X \rightarrow X$ is *idempotent* if $f \circ f = f$, i.e., $f(f(x)) = f(x)$, $\forall x \in X$. Show that if f is continuous and idempotent on a Hausdorff space X , then the image $f(X)$ is closed. *Hint:* show that if $w \in \overline{f(X)}$ then $f(w) = w$.
4. Let $X = \prod_{\alpha} X_{\alpha}$ be an infinite product with the usual product topology. Show that if $\emptyset \neq A_{\alpha} \subseteq X_{\alpha}$ then the closure

$$\overline{\prod_{\alpha} A_{\alpha}} = \prod_{\alpha} \overline{A_{\alpha}}$$

5. Show that a continuous mapping $f : X \rightarrow X$ on a compact Hausdorff space X has a fixed set, i.e., $\emptyset \neq A \subseteq X$ such that $f(A) = A$.
Hint: Use the finite intersection property.